

Write a program to reverse an integer in Python.

- **Example:**
 - Input: 1234
 - Output: 4321

Write a program in Python to check whether an integer is an Armstrong number or not.

- **Example:**
 - Input: 153
 - Output: True

Write a program in Python to check if a given number is prime or not.

- **Example:**
 - Input: 17
 - Output: True

Write a program in Python to print the Fibonacci series using an iterative method.

- **Example:**
 - Input: 5
 - Output: [0, 1, 1, 2, 3]

Write a program in Python to print the Fibonacci series using a recursive method.

- **Example:**
 - Input: 5
 - Output: [0, 1, 1, 2, 3]

Write a program in Python to check whether a number is a palindrome or not using an iterative method.

- **Example:**
 - Input: 121

- Output: True

Write a program in Python to check whether a number is a palindrome or not using a recursive method.

- **Example:**

- Input: 121
- Output: True

Write a program in Python to find the greatest among three integers.

- **Example:**

- Input: 10, 20, 30
- Output: 30

Write a program in Python to check if a number is binary.

- **Example:**

- Input: 1010
- Output: True

Write a program in Python to find the sum of digits of a number using recursion.

- **Example:**

- Input: 1234
- Output: 10

Write a program in Python to swap two numbers without using a third variable.

- **Example:**

- Input: a = 5, b = 10
- Output: a = 10, b = 5

Write a program in Python to swap two numbers using a third variable.

- **Example:**

- Input: a = 5, b = 10

- Output: a = 10, b = 5

Write a program in Python to find prime factors of a given integer.

- **Example:**

- Input: 28
- Output: [2, 7]

Write a program in Python to add two integers without using arithmetic operators.

- **Example:**

- Input: 5, 3
- Output: 8

Write a program in Python to find if a given number is perfect or not.

- **Example:**

- Input: 28
- Output: True

Write a Python program to find the average of numbers.

- **Example:**

- Input: [10, 20, 30]
- Output: 20.0

Write a Python program to calculate the factorial using an iterative method.

- **Example:**

- Input: 5
- Output: 120

Write a Python program to calculate the factorial using recursion.

- **Example:**

- Input: 5
- Output: 120

Write a Python program to check if a given number is even or odd.

- **Example:**
 - Input: 4
 - Output: Even

Write a Python program to print the first n prime numbers with an explanation.

- **Example:**
 - Input: 5
 - Output: [2, 3, 5, 7, 11]

Write a Python program to print prime numbers in a given range.

- **Example:**
 - Input: 10 to 30
 - Output: [11, 13, 17, 19, 23, 29]

Write a Python program to find the smallest number among three.

- **Example:**
 - Input: 10, 20, 30
 - Output: 10

Write a Python program to convert a decimal number into binary.

- **Example:**
 - Input: 10
 - Output: 1010

Write a Python program to convert a decimal number to octal.

- **Example:**
 - Input: 8
 - Output: 10

Write a Python program to check if the given year is a leap year or not.

- **Example:**
 - Input: 2020
 - Output: True

Write a Python program to calculate simple interest.

- **Example:**
 - Input: Principal = 1000, Rate = 5%, Time = 2
 - Output: 100.0

Write a Python program to remove a given character from a string.

- **Example:**
 - Input: "hello", 'l'
 - Output: "heo"

Write a Python program to count occurrences of a given character in a string.

- **Example:**
 - Input: "hello", 'l'
 - Output: 2

Write a Python program to check if two strings are anagrams.

- **Example:**
 - Input: "listen", "silent"
 - Output: True

Write a Python program to check if a string is a palindrome.

- **Example:**
 - Input: "radar"
 - Output: True

Write a Python program to check if a given character is a vowel or consonant.

- **Example:**
 - Input: 'a'
 - Output: Vowel

Write a Python program to check if a given character is a digit or not.

- **Example:**
 - Input: '5'
 - Output: True

Write a Python program to replace spaces in a string with a given character.

- **Example:**
 - Input: "hello world", '-'
 - Output: "hello-world"

Write a Python program to convert lowercase characters to uppercase in a string.

- **Example:**
 - Input: "hello"
 - Output: "HELLO"

Write a Python program to convert lowercase vowels to uppercase in a string.

- **Example:**
 - Input: "hello"
 - Output: "hEllo"

Write a Python program to delete vowels in a given string.

- **Example:**
 - Input: "hello"
 - Output: "hll"

Write a Python program to count the occurrences of vowels and consonants in a string.

- **Example:**
 - Input: "hello"
 - Output: Vowels: 2, Consonants: 3

Write a Python program to print the highest frequency character in a string.

- **Example:**
 - Input: "hello"
 - Output: 'l'

Write a Python program to replace the first occurrence of a vowel with '-' in a string.

- **Example:**
 - Input: "hello"
 - Output: "h-llo"

Write a Python program to count alphabets, digits, and special characters in a string.

- **Example:**
 - Input: "Hello123!@#"
 - Output: Alphabets: 5, Digits: 3, Special Characters: 3

Write a Python program to separate characters in a given string.

- **Example:**
 - Input: "hello"
 - Output: ['h', 'e', 'l', 'l', 'o']

Write a Python program to remove blank spaces from a string.

- **Example:**
 - Input: " hello world "
 - Output: "helloworld"

Write a Python program to concatenate two strings using the join() method.

- **Example:**
 - Input: "Hello", "World"
 - Output: "Hello World"

Write a Python program to concatenate two strings without using the join() method.

- **Example:**
 - Input: "Hello", "World"
 - Output: "HelloWorld"

Write a Python program to remove repeated characters from a string.

- **Example:**
 - Input: "hello"
 - Output: "helo"

Write a Python program to calculate the sum of integers in a string.

- **Example:**
 - Input: "abc123"
 - Output: 6

Write a Python program to print all non-repeating characters in a string.

- **Example:**
 - Input: "hello"
 - Output: 'h', 'e', 'o'

Write a Python program to copy one string to another string.

- **Example:**
 - Input: "hello"
 - Output: "hello" (copied)

Write a Python program to sort characters of a string in ascending order.

- **Example:**
 - Input: "dcba"
 - Output: "abcd"

Write a Python program to sort characters of a string in descending order.

- **Example:**
 - Input: "dcba"
 - Output: "dcba"

Write a Python program to convert a binary number to decimal.

- **Example:**
 - Input: 1010
 - Output: 10

Write a Python program to find the second largest number in a list.

- **Example:**
 - Input: [1, 2, 3, 4, 5]
 - Output: 4

Write a Python program to find the intersection of two lists.

- **Example:**
 - Input: [1, 2, 3], [2, 3, 4]
 - Output: [2, 3]

Write a Python program to find the union of two sets.

- **Example:**
 - Input: {1, 2, 3}, {3, 4, 5}
 - Output: {1, 2, 3, 4, 5}

Write a Python program to calculate the length of a string without using the len() function.

- **Example:**
 - Input: "hello"
 - Output: 5

Write a Python program to print the first n Fibonacci numbers using memoization.

- **Example:**
 - Input: 5
 - Output: [0, 1, 1, 2, 3]

Write a Python program to check if a string contains only digits.

- **Example:**
 - Input: "1234"
 - Output: True

Write a Python program to find the most common element in a list.

- **Example:**
 - Input: [1, 2, 2, 3, 3, 3]
 - Output: 3

Write a Python program to find the longest word in a sentence.

- **Example:**
 - Input: "The quick brown fox"
 - Output: "quick"

Write a Python program to check if a string is a rotation of another string.

- **Example:**
 - Input: "abcd", "dabc"
 - Output: True

Write a Python program to implement a basic calculator.

- **Example:**
 - Input: 2 + 3
 - Output: 5

Write a Python program to find the minimum and maximum number in a list without using built-in functions.

- **Example:**
 - Input: [10, 20, 5, 30]
 - Output: Min: 5, Max: 30

Write a Python program to print a pyramid pattern of stars.

- **Example:**
 - Input: 3

Output:

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*  
***  
*****
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Write a Python program to find all pairs in a list that sum up to a specific target.

- **Example:**
 - Input: [1, 2, 3, 4, 5], 5
 - Output: [(1, 4), (2, 3)]

Write a Python program to find the first non-repeating character in a string.

- **Example:**
 - Input: "swiss"
 - Output: 'w'

Write a Python program to calculate the area of a circle.

- **Example:**
 - Input: radius = 5
 - Output: 78.54

Write a Python program to merge two dictionaries.

- **Example:**
 - Input: {'a': 1}, {'b': 2}
 - Output: {'a': 1, 'b': 2}

Write a Python program to find the index of an element in a list.

- **Example:**
 - Input: [1, 2, 3, 4], 3
 - Output: 2

Write a Python program to create a countdown timer.

- **Example:**
 - Input: 5 (countdown from 5)
 - Output: 4, 3, 2, 1, 0

Write a Python program to generate a random password.

- **Example:**
 - Output: gY@9eR*

Write a Python program to remove duplicates from a list while maintaining order.

- **Example:**
 - Input: [1, 2, 2, 3, 4, 4]
 - Output: [1, 2, 3, 4]

Write a Python program to check if a given string is a valid email address.

- **Example:**
 - Input: "example@domain.com"
 - Output: True

Write a Python program to calculate the power of a number using recursion.

- **Example:**
 - Input: 2, 3
 - Output: 8

Write a Python program to implement a queue using lists.

- **Example:**
 - Input: enqueue(5), enqueue(10), dequeue()
 - Output: 5

Write a Python program to implement a stack using lists.

- **Example:**

- Input: push(5), push(10), pop()
- Output: 10

Write a Python program to print the ASCII value of a character.

- **Example:**

- Input: 'A'
- Output: 65

Write a Python program to find the intersection of two arrays.

- **Example:**

- Input: [1, 2, 3], [2, 3, 4]
- Output: [2, 3]

Write a Python program to convert a string to a list of words.

- **Example:**

- Input: "Hello World"
- Output: ['Hello', 'World']

Write a Python program to implement binary search.

- **Example:**

- Input: [1, 2, 3, 4, 5], 3
- Output: 2 (index of the number 3)

Write a Python program to find all permutations of a string.

- **Example:**

- Input: "abc"
- Output: ['abc', 'acb', 'bac', 'bca', 'cab', 'cba']

Write a Python program to find the longest common prefix in a list of strings.

- **Example:**

- Input: ["flower", "flow", "flight"]
- Output: "fl"

Write a Python program to find the maximum product of two integers in a list.

- **Example:**

- Input: [1, 2, 3, 4]
- Output: 12 (3 * 4)

Write a Python program to rotate an array to the right by k steps.

- **Example:**

- Input: [1, 2, 3, 4, 5], 2
- Output: [4, 5, 1, 2, 3]

Write a Python program to implement the sieve of Eratosthenes to find all prime numbers up to n.

- **Example:**

- Input: 30
- Output: [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

Write a Python program to implement a basic text-based Tic-Tac-Toe game.

- **Example:**

- Input: Players take turns entering their moves.
- Output: Display the game board and announce the winner.

Write a Python program to create a simple URL shortener.

- **Example:**

- Input: "https://www.example.com"
- Output: Shortened URL: "http://short.ly/abcd"

Write a Python program to check for balanced parentheses in an expression.

- **Example:**

- Input: "((a + b) * (c + d))"
- Output: True

Write a Python program to serialize and deserialize a binary tree.

- **Example:**

- Input: A binary tree structure.
- Output: Serialized string representation of the tree.

Write a Python program to find the nth Fibonacci number using memoization.

- **Example:**

- Input: 10
- Output: 55

Write a Python program to flatten a nested list.

- **Example:**

- Input: [1, [2, [3, 4]], 5]
- Output: [1, 2, 3, 4, 5]

Write a Python program to implement a simple stack with push and pop operations.

- **Example:**

- Input: push(10), push(20), pop()
- Output: 20

Write a Python program to find the frequency of each character in a string using a dictionary.

- **Example:**

- Input: "hello"
- Output: {'h': 1, 'e': 1, 'l': 2, 'o': 1}

Write a Python program to find the first repeating character in a string.

- **Example:**

- Input: "geeksforge"
- Output: 'g'

Write a Python program to count the number of words in a string.

- **Example:**
 - Input: "Hello World"
 - Output: 2

Write a Python program to check if two strings are rotations of each other.

- **Example:**
 - Input: "abcd", "cdab"
 - Output: True

Write a Python program to implement a queue using collections.deque.

- **Example:**
 - Input: enqueue(1), enqueue(2), dequeue()
 - Output: 1

Write a Python program to implement depth-first search (DFS) on a graph.

- **Example:**
 - Input: A graph represented as an adjacency list.
 - Output: Traversal order of nodes.

Write a Python program to generate all subsets of a set. - Example:

- Input: {1, 2}
- Output: [], [1], [2], [1, 2]